Zhitong Chen

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Education

Texas A&M University — College Station, TX
Ph.D. Student in Computer Science, advised by Prof. **Ali Mostafavi** and **James Caverlee** — Jan. 2026 – Expected May/Aug 2028

Texas A&M University — College Station, TX
M.S. in Computer Science, advised by Prof. **Ali Mostafavi** — Aug. 2023 – Dec. 2025

University of Liverpool — UK
B.Sc. in Information and Computing Science (Computer Science) — Sept. 2018 – June 2022

Research Interests

LLM Agents, Structured Tool Planning, Test-Time Inference, Agent Learning and Post-Training, RAG, and LLM Evaluation.

Research Experience

Agentic Learning and LLM Post-Training for Tool Planning *(Ongoing)*
- Investigating learning from successful and failed search trajectories to improve direct structured workflow generation; analyzing training signals and generalization to unseen tool combinations.

Test-Time Inference and Candidate Selection for LLM Planning **[code]**
- Compared Monte Carlo Tree Search (MCTS), best-of-N sampling, and multi-operator search for structured tool-use planning.
- Introduced a Discovery–Validation–Selection (DVS) decomposition and prompt-matched controls; found that apparent search advantages can disappear under controlled comparisons while correct workflows remain unselected.

DisasterBench: Structured Tool Planning Benchmark **[code]**
- Developed an expert-validated benchmark of **233 tasks**, **26 tools**, and **81 typed dependencies** for LLM workflow planning in disaster response.
- Introduced First-Point-of-Failure (FPoF) analysis to diagnose failures in tool choice, parameters, dependencies, and agent-workflow alignment.

Harvey Streamflow and Community Impact Forecasting Benchmark *(Ongoing)*
- Developing time-aware SSF/CIF benchmark evaluations using historical observations, forecast horizons, and baselines; examining label validity, information leakage, and forecastability.

DisastQA: Domain-Specific LLM and RAG Evaluation **[code]**
- Developed **3,000 verified questions** across **8 disaster types**, covering multiple-choice and open-ended question answering.
- Evaluated **20 LLMs** under varying evidence conditions, including keypoint-based evaluation of open-ended answers.

Texas Disaster Information System (TDIS): RAG Evaluation
- Evaluated **1.7M+ retrieval-generation instances** across retrieval, reranking, and LLM settings; investigated multi-source disaster information workflows.

Selected Publications and Manuscripts

DisastQA: A Comprehensive Benchmark for Evaluating Question Answering in Disaster Management
Zhitong Chen*, Kai Yin*, Xiangjue Dong, Chengkai Liu, Xiangpeng Li, Bo Li, Junwei Ma, Yiming Xiao, Ali Mostafavi, James Caverlee
Findings of ACL 2026 [paper] [code]

DisasterBench: Benchmarking LLM Planning under Typed Tool Interface Constraints
Zhitong Chen*, Kai Yin*, Weifeng Zhang, Zhiyuan Wang, Xiangjue Dong, Chengkai Liu, Zhewei Liu, Yiming Xiao, Ali Mostafavi, James Caverlee
Findings of EMNLP 2026 [paper] [code]

From Apparent Search Gains to Selection Failures: Auditing Test-Time Inference in Structured Workflow Planning
Zhitong Chen, Weifeng Zhang, Kai Yin, Zhiyuan Wang, Ali Mostafavi, James Caverlee
arXiv preprint, 2026; Under review at ACL Rolling Review [paper] [code]

FloodSQL-Bench: A Retrieval-Augmented Benchmark for Geospatially-Grounded Text-to-SQL
Hanzhou Liu, Kai Yin, **Zhitong Chen**, Chenyue Liu, Ali Mostafavi
AAAI Summer Symposium 2026 [paper]

DisastRAG: A Multi-Source Disaster Information Integration and Access System Based on Retrieval-Augmented Large Language Models
Bo Li, **Zhitong Chen**, Kai Yin, Junwei Ma, Yiming Xiao, Ali Mostafavi
arXiv preprint, 2026 [preprint]

* Equal contribution.

Academic Service

Reviewer, NeurIPS 2026 (4 papers). Poster Presenter, ACL 2026 (San Diego, CA).

Technical Skills

Programming / ML: Python, SQL, PyTorch, Hugging Face Transformers, vLLM, LangChain.
Methods / Tools: LLM evaluation, MCTS, RAG, FAISS, BM25, pandas, NumPy, Git, Linux.